

Lukas A. White

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EDUCATION

Saginaw Valley State University

University Center, MI

Bachelor of Science in Computer Science — Anticipated May 2026 | GPA: 3.52 / 4.0

- Dean's List: Winter 2023, Fall 2024 | University Honors: 2024

ACTIVITIES

Pledge of the Computing Professional

- Acknowledged that my work impacts individuals and society, both now and in the future. I commit to uphold ethical principles, act with integrity and respect, and use my skills responsibly and for the public good.
- Resolved to remain honest about my capabilities, pursue continual learning, and conduct myself with professionalism in all aspects of my work.

CS Capstone Project — QA Technical Lead

In Progress

C#, SQL, Software Quality Assurance, Test Planning, Database Design, Windows Desktop Application

- Leading QA efforts for a nonprofit pet foster management system built for Midland County Pit Stop, a rescue organization that has placed thousands of animals since 2018.
- Defining and executing test plans for core modules including pet profiles, adoption records, foster/volunteer tracking, supply requests, calendar/reminders, and financial reporting (revenue, expenses, 990 form support).
- Validating role-based access controls, SQL data integrity, contract generation, and financial recordkeeping against real-world client requirements.

AI Task Force — Saginaw Valley State University

Python, AI/LLMs, Automation, Systems Programming, Research & Prototyping

- Researched and experimented with emerging AI capabilities, focusing on pushing the practical and technical limits of modern AI tools.
- Developed unconventional systems including a FORTRAN-based database server and a custom compiler and virtual machine that computes target values using constrained instruction sets and register operations.
- Explored AI-assisted automation using Microsoft Power Automate, including workflows for generating and sending AI-driven communications.
- Evaluated and prototyped with modern AI platforms such as Replit and Synthesia for code generation and AI-based training video creation.
- Conducted rapid prototyping across multiple technologies, demonstrating adaptability and the ability to learn unfamiliar tools quickly.
- Documented findings and contributed insights on real-world applications and limitations of current AI systems.

Game Design Club — SVSU

March 2025 – Present

- Contributed original gameplay ideas and design approaches during collaborative brainstorming sessions.
- Evaluated feasibility of proposed concepts and guided peers in turning ideas into working prototypes.
- Demonstrated practical methods for overcoming technical and creative design challenges through collaborative problem-solving.

SAE Formula Racing Team — SVSU

January 2022 – January 2024

- Designed and programmed a lap-timing system in Arduino/C++, improving team precision in time-critical race scenarios.
- Collaborated with teammates to troubleshoot complex electrical and software integration issues.

PROJECTS

Independent Study: AI Agent Development

In Progress

Python, AI Fundamentals, Data Preprocessing, GitHub, LaTeX (Overleaf), Research Writing, Public Speaking

- Surveyed modern AI agent frameworks and synthesized findings into a structured technical report.
- Built experimental agents in Python with Gemini API integration, covering data preprocessing and version control via GitHub.
- Delivered a two-day presentation teaching classmates to implement agents using both no-code and code-based AI development platforms.

Web-Based Study Tool

Group Project

HTML, CSS, JavaScript, Node.js, Express, MongoDB, AWS (EC2, S3, Route53)

- Led a four-person team, mentoring peers in Node.js, Express, and database integration while managing scope and delivery timelines.
- Built a full-stack academic productivity platform featuring a word processor, flashcards with text-to-speech, drag-and-drop diagrams, JWT authentication, CRUD operations, and MongoDB persistence.
- Ensured cross-browser compatibility and responsive design for a seamless experience on desktop and mobile.

Client-Server Sales System

Class Project

Java, JavaFX, SQLite, Multi-Threading, Network Programming, Load Balancing

- Built a multi-threaded client-server application supporting 1,000+ concurrent transactions with real-time communication, error handling, and data serialization.
- Implemented SQLite database logic and SQL queries to ensure reliable, high-performance sales data storage and integrity.
- Designed a scalable architecture with load balancing across server threads and a JavaFX GUI for product entry, receipt generation, and sales tracking.

Lap Timing System — Formula SAE

Personal / Team Project

Arduino, Embedded C, Sensors, Signal Handling, Electrical Engineering Fundamentals

- Designed and built a custom lap-timing system using an Arduino and garage-door sensors to detect vehicle pass events in real time.
- Integrated hardware components and applied foundational electrical engineering concepts, including signal handling and sensor interfacing.
- Diagnosed hardware and display limitations and implemented a fallback solution routing serial output directly to a console for reliable monitoring.
- Contributed to iterative testing and system validation within a multidisciplinary engineering environment.

Turbo Auto Service Scheduling System

Class Project — CS 421

Java, SQLite, File I/O, Database Design, CLI Development

- Built a scheduling and database-driven application to manage automotive service workflows across multiple mechanics.
- Designed a SQLite-backed system for storing customers, vehicles, services, and scheduling data with full CRUD support.

- Developed file import functionality to parse structured text datasets and populate relational database tables.
- Implemented dynamic scheduling logic to distribute workload and calculate time and pay across day, week, month, and year timeframes.
- Engineered database reset and deletion features with user confirmation safeguards to prevent accidental data loss.

MDI Number System Calculator

Class Project — CIS 311

C#, .NET WinForms, Object-Oriented Programming, Bitwise Operations, UI/UX Design

- Developed a multi-document interface (MDI) desktop application supporting multiple simultaneous calculator windows.
- Implemented real-time conversion between binary, decimal, and hexadecimal number systems using custom parsing and validation logic.
- Designed interactive UI controls that dynamically restrict valid input based on the selected number base.
- Built bitwise logic operations (AND, OR, XOR, NOT) and displayed results across multiple number formats.

Open-World Island Simulation — Wii Homebrew

Personal Project

C, GRRLIB, DevKitPro, GCCORE, Procedural Generation, Spatial Algorithms

- Engineered a semi-infinite, procedurally generated 3D world featuring ocean and island terrain.
- Developed custom despawning logic with real-time collision detection and kd-tree spatial queries, optimizing memory usage for seamless exploration.
- Optimized simulation performance to run smoothly within the hardware constraints of the Wii platform.

Simple Machine Learning Simulation in Unity

Personal Project

Python, C#, Unity, Reinforcement Learning

- Created an agent that learns via trial-and-error to locate a target object within a 3D environment.
- Integrated Python and C# scripts to control learning logic and environment interaction.
- Demonstrated the ability to apply machine learning concepts within an interactive simulation.

SKILLS

Programming Languages: Python (primary), C/C++, Java, C#, JavaScript (ES6+), Visual Basic

Web Development & Cloud: HTML5, CSS3, Node.js, Express.js, MongoDB, AWS (EC2, S3, Route53), RESTful API Design, JWT Authentication

Databases & Systems: SQL, SQLite, NoSQL (MongoDB), Database Design, Client-Server Architecture, Load Balancing, File I/O

AI & Machine Learning: AI Agent Development, Gemini API, Reinforcement Learning, Data Preprocessing, ML Fundamentals, Po

Embedded & Systems Programming: Arduino, Embedded C, Signal Handling, Sensor Integration, Procedural Generation, Spatial Algorithms (kd-tree)

Software & Tools: Git & GitHub (branching, pull requests, code review), Unity, Visual Studio, VS Code, LaTeX (Overleaf), Microsoft Office, Arduino IDE

Core CS Competencies: Object-Oriented Programming, Multi-threaded Programming, Network Programming, API Integration, Algorithm Design, UI/UX Design

Professional Skills: Team Leadership, Technical Mentorship, Project Management, Research & Technical Writing, Public Speaking, Problem Solving

WORK EXPERIENCE

Subway*Auburn, MI | March 2025 – Present***Staff**

- Trained and onboarded new employees, ensuring consistent performance and smooth team integration.
- Delivered fast, friendly service in a high-volume environment, serving 200+ customers daily.
- Maintained cleanliness, safety, and quality standards, contributing to a 99/100 health inspection score.

Midland Civic Arena*Midland, MI | June 2021 – December 2021***Seasonal Staff**

- Managed 250+ daily guest interactions, ensuring smooth operations and compliance with safety protocols.
- Trained and mentored new employees, supporting team efficiency and consistently high service quality.
- Assisted with event logistics and on-the-spot problem-solving to enhance the overall visitor experience.